

Current-limited capability of grid-forming converters in reserve and inertia scheduling: formulation, sign condition and validation

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Abstract

Scheduling formulations in which energy, reserve and inertia share the capability of a grid-forming converter bound that capability by active power and omit the current limit that couples the three services. This paper represents the continuous and short-term current ratings, reactive current and terminal voltage of the converter as a polyhedral capability set and integrates it in a multi-period linear clearing whose prices are dual variables. A piecewise condition is derived for the sign of the error made by the active-power bound; it depends on the short-term current radius, the bound and the reactive current held during the response. Final schedules on the IEEE 39-bus and RTS-24 systems are obtained by successive linearisation and satisfy bus voltage, generator reactive, branch loading and balance limits in an ac power flow for both representations. The procured response is constrained to be deliverable after the largest generator outages, and a current-limited RMS converter model confirms that the scheduled inertial power is delivered within the current limit at the terminal voltage reached during the event. The three omissions change the cleared cost with different signs, so a constant derating cannot correct the bound. Across four rating conventions, two bound definitions, three comparison models and 605 randomised converter placements, the capability set is never the more expensive representation. The cost difference is 0.3 to 1.1 % at the base case and depends on the rating convention; the inertia price under the active-power bound is 2.3 to 13.7 times that under the capability set.

Keywords: grid-forming converter, inertia, operating reserve, capability curve, ancillary service pricing, converter current limit

1. Introduction

As synchronous generation is displaced, inertia and fast reserve have to be procured explicitly, and increasingly from converter-interfaced resources [1, 2, 3]. Grid-forming (GFM) control, in which the converter behaves as a voltage source behind an impedance, is the main technical response [4, 5, 6], and system operators now specify GFM capability rather than encourage it [7, 8, 9]. The energy a GFM converter sells, the reserve it holds and the inertial power it injects after a disturbance all pass through the same semiconductor bridge, so they are limited jointly, and the limit is a current [10].

Scheduling formulations have treated the frequency side of this problem in increasing detail. Governor rate limits entered the optimal power flow in [11]; inertia became a decision variable in [12]; frequency services were co-optimised in [13, 14]; and requirement uncertainty and the siting of virtual inertia were addressed in [15, 16]. On the supply side these models bound what a resource can offer by active power. Three recent formulations require converter inertia to be backed by unused headroom [17, 18, 19], and a fourth embeds a learned frequency surrogate in the day-ahead schedule [20]. They differ in most respects, but in each the converter capability has the form

$$P_i + R_i + H_i \leq \bar{P}_i, \quad (1)$$

with P_i the energy schedule, R_i the reserve and H_i the inertial power. None contains a current limit, a reactive-power term or the terminal voltage (Table 1).

System operators describe the capability differently. AEMO specifies overload capability as a level with a duration, for example 150 % of rated current for 2 s, and reports that the active headroom of an inverter depends on its pre-disturbance operating point [21, 22]; MISO's draft specification addresses operation at short-duration current limits [23]. At device and feeder level, reactive support has been shown to erode the active reserve available for fast frequency response, through the inverter losses at the operating point on its capability curve [24], and current-based inverter limits have been used in photovoltaic-storage scheduling and in batteries providing frequency and voltage control together [25, 26]. The remaining gap is a market clearing in which the current limit is shared between energy, reserve and inertia, the requirement multipliers are prices, and the effect of the representation is quantified.

The physical limit on a converter is

$$\sqrt{P^2 + Q^2} \leq V I_{\max} S, \quad (2)$$

a disc in the P - Q plane whose radius scales with the terminal voltage V . Equation (1) is the projection of Eq. (2) onto $Q = 0$ at $V = 1.0$ pu with $I_{\max}$ fixed at the continuous rating. It therefore omits the short-term current rating, which understates the capability available for an inertial response; omits reactive current, which overstates capability whenever $Q \neq 0$; and fixes

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Nomenclature

i, g, t, z, ω	converter, synchronous unit, period, zone, outage	$P_i, Q_i, \tilde{Q}_i$	active, reactive, short-term reactive power
$S_i, P_i^{\text{sch}}, \bar{P}_i$	rating, scheduled active bound, comparison bound	R_i, H_i	reserve, inertial power (MW)
I^c, I^s	continuous, short-term current limit (pu)	p^d, p^c, c, e	discharge, charge, curtailment, available injection
V_i, V_i^{sh}	pre-event, event terminal voltage (pu)	a_i, σ_i	band availability, state of charge
α_i, χ	reactive priority, reserve overlap	p^g, r^g	synchronous energy, reserve
m, β_i, L_i	polygon faces, band width, short-term radius	λ, μ	energy price, requirement prices
σ^e, σ^*, E_i	energy floor, full-band breakpoint, store energy	$\rho, \varphi, f^{\text{max}}$	reserve, reactive requirement fractions, RoCoF limit
τ^R, τ^H	reserve, inertia delivery windows	$\Psi, \mathbf{f}^{\text{max}}, \varepsilon$	shift factors, branch ratings, emergency factor
$\mathcal{H}_g, S_g$	inertia constant (s), rating of unit g	$\delta, \tilde{Q}^{\text{crit}}$	cost difference, crossing of the Proposition
$\kappa^\bullet$	offers	x_i^g, ϕ_ω^t	reactance behind terminal, deployed fraction
$\gamma^{\text{UD}}, \kappa^Q$	uniform derating factor, reactive range	$\mathcal{I}, \mathcal{G}, \mathcal{Z}_I$	converters, synchronous units, zones with converters

Table 1: Scheduling formulations with shared converter capability and what each represents ($\bullet$ present, $\circ$ absent, — not applicable). Entries follow the constraint sets stated in each paper.

Reference	Test system	Shared capability	Current limit	Reactive power	Terminal voltage	Short-term $\neq$ continuous	Storage state	Prices as duals	Frequency model
Park et al. [17]	IEEE 30	$\bullet$	$\circ$	$\circ$	$\circ$	$\circ$	$\bullet$	$\bullet$	Analytic nadir
Jiang and Li [18]	WSCC 9, IEEE 39	$\bullet$	$\circ$	$\circ$	$\circ$	$\circ$	$\circ$	$\circ$	EMT surrogate
Jiang and Li [20]	IEEE 9	$\bullet$	$\circ$	$\circ$	$\circ$	$\circ$	$\bullet$	$\circ$	EMT surrogate
Beigi Fard et al. [19]	IEEE 9, 118	$\bullet$	$\circ$	$\circ$	$\circ$	$\circ$	$\bullet$	$\circ$	PWL nadir
Garozzo and Tina [24]	inverter, LV feeder	—	$\circ$	$\bullet$	$\circ$	$\circ$	$\bullet$	—	—
This work	IEEE 39, RTS-24	$\bullet$	$\bullet$	$\bullet$	$\bullet$	$\bullet$	$\bullet$	$\bullet$	RMS check

the voltage, which overstates capability below 1.0 pu and understates it above. The three errors do not have the same sign, so a constant derating of Eq. (1) cannot remove them together.

The contributions of this paper are as follows.

1. The continuous and short-term current ratings, reactive current and terminal voltage of a GFM converter are represented as a polyhedral capability set and integrated in a multi-period joint clearing of energy, reserve and inertia with marginal pricing. The clearing remains a linear programme, so the prices are dual variables.
2. A piecewise condition is derived for the sign of the error made by the active-power bound of Eq. (1). The condition depends on the short-term radius, the bound and the reactive current held during the response, and not on the sustained active power.
3. The formulation is evaluated on the IEEE 39-bus and RTS-24 systems against three comparison models and an exact second-order cone benchmark, under alternative converter rating conventions and randomised converter placements. The final schedules are validated by ac power flow against voltage, generator reactive, branch and balance limits; the procured response is constrained to be deliverable after the largest generator outages; and the scheduled inertial power is checked against a current-limited RMS converter model.

Fig. 1 summarises the approach. The formulation extends the shared-capability clearing of [17], whose treatment of inertia as a priced power service and of storage energy recovery is retained; what changes is the shape of the set the services share.

2. Capability set and sign condition

2.1. Continuous and short-term limits

Converter i has apparent-power rating S_i , terminal voltage V_i (pu), continuous current limit I^c and short-term current limit $I^s \geq I^c$ (pu of rated current). Reserve that is called must be sustained for the containment period, whereas the inertial response lasts of the order of a second. Two constraints follow:

$$\sqrt{(P_i + R_i)^2 + Q_i^2} \leq V_i I^c S_i, \quad (3)$$

$$\sqrt{(P_i + \chi R_i + H_i)^2 + \tilde{Q}_i^2} \leq V_i^{\text{sh}} [I^c + a_i(I^s - I^c)] S_i. \quad (4)$$

Equation (3) applies the continuous (thermal) rating to the sustained schedule. It is imposed at both ends of the reserve activation, (P_i, Q_i) and $(P_i + R_i, Q_i)$, because a charging converter ($P_i < 0$) carries its largest current before its reserve is called; the set is convex, so the intermediate points are covered. Equation (4) applies the short-term rating at the instant of the inertial response. The factor $\chi \in [0, 1]$ is the fraction of the reserve already activated at that instant and is set to one, the simultaneous envelope. V_i^{sh} is the terminal voltage during the event; it equals V_i unless the correction of Section 5.6 is applied. The band availability $a_i \in [0, 1]$ is defined below. Fig. 2 compares the two constraints with Eq. (1).

2.2. Reactive current priority

During a short-term excursion the current-priority setting of the converter determines how much reactive current is retained [27, 28]. The setting is a property of the converter, so it enters as a parameter:

$$\tilde{Q}_i = \alpha_i Q_i, \quad \alpha_i \in [0, 1], \quad (5)$$

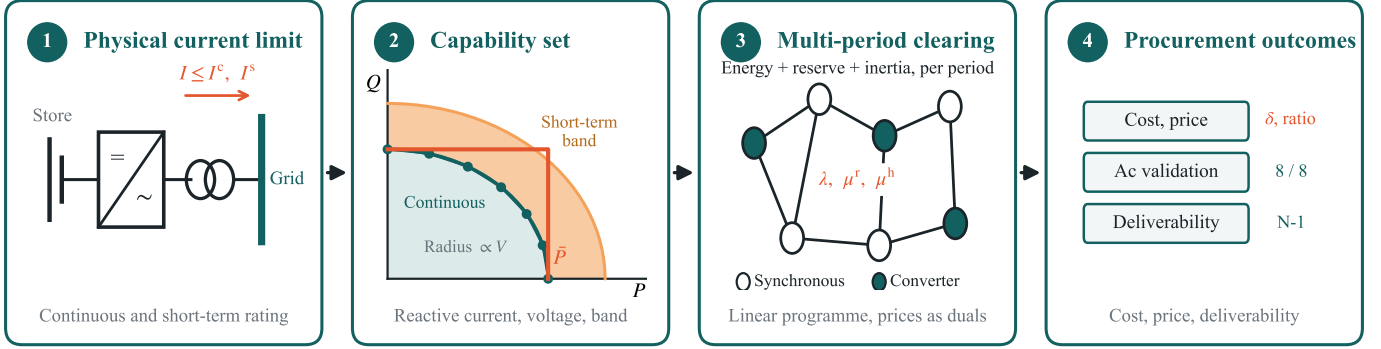


Figure 1: Overview: physical current limit, polyhedral capability set, multi-period clearing with network and ac validation, and procurement outcomes.

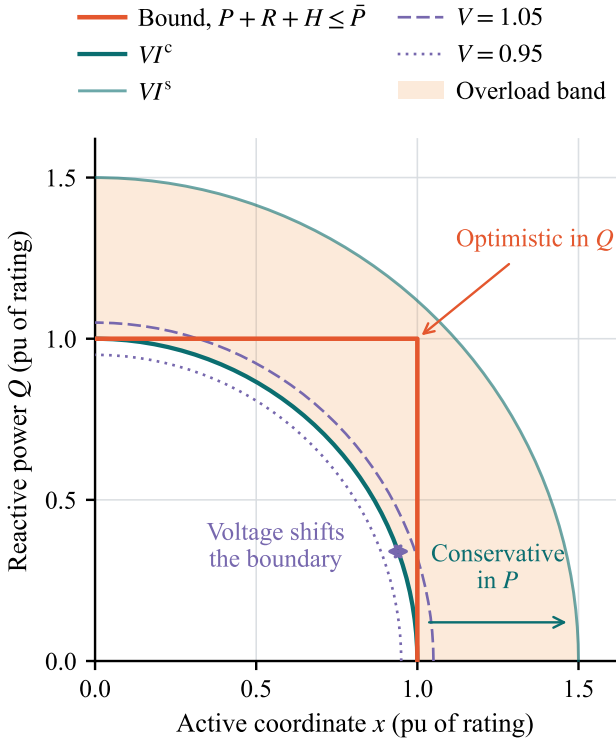


Figure 2: Capability set and active-power bound at $I^c = 1.0$ pu, $I^s = 1.5$ pu.

with $\alpha_i = 1$ for reactive priority and $\alpha_i = 0$ for ideal active priority. Writing the relation as a bound $0 \leq \tilde{Q}_i \leq Q_i$ would let the optimisation choose the converter's current allocation.

2.3. Storage state of charge

A store limits the services in two ways. The short-term band is available only above an energy floor $\underline{\sigma}^e$ and fully available above a breakpoint σ^* ,

$$0 \leq a_i^t \leq \min\left(1, \frac{\min(\sigma_i^{t-1}, \sigma_i^t) - \underline{\sigma}^e}{\sigma^* - \underline{\sigma}^e}\right), \quad (6)$$

and the energy behind the services must be held for their delivery windows τ^R and τ^H ,

$$\frac{\tau^R R_i^t + \tau^H H_i^t}{\eta_d} \leq (\min(\sigma_i^{t-1}, \sigma_i^t) - \underline{\sigma}^e) E_i. \quad (7)$$

Both are evaluated at the two ends of the period and are linear in the state of charge. The linear ramp in Eq. (6) is an assumption; moving the breakpoint σ^* across its range changes the reported cost differences by less than 0.05 percentage points.

2.4. Polyhedral form

Equations (3) and (4) are second-order cones. Each is replaced by an inscribed regular m -gon [29],

$$(P_i + R_i) \cos \theta_k + Q_i \sin \theta_k \leq V_i I^c S_i \cos \frac{\pi}{m}, \quad (8)$$

$$(P_i + \chi R_i + H_i) \cos \theta_k + \tilde{Q}_i \sin \theta_k - a_i \beta_i \leq V_i^{\text{sh}} I^c S_i \cos \frac{\pi}{m}, \quad (9)$$

for $\theta_k = 2\pi k/m$, $k = 0, \dots, m-1$, with $\beta_i = V_i^{\text{sh}}(I^s - I^c) S_i \cos(\pi/m)$ the width of the overload band. The band width is a parameter, so both constraints are affine. The polygon is an inner approximation: the clearing cost is an upper bound on that of the exact conic problem. With $m = 24$ the cleared cost exceeds the exact second-order cone solution by at most 0.066 % over the configurations of Section 5, and by 0.0045 % with $m = 96$. The case study uses $m = 24$, and $m = 192$ where the polygon error would be comparable to the effect measured.

2.5. Sign of the error of the active-power bound

Consider a sustained operating point admitted by both representations. The radius allowed by the short-term constraint is

$$L_i = V_i^{\text{sh}} [I^c + a_i(I^s - I^c)] S_i. \quad (10)$$

The active-power bound leaves inertial headroom $H_i^{\text{ap}} = \bar{P}_i - (P_i + R_i)$ and the capability set leaves $H_i^{\text{set}} = \sqrt{L_i^2 - \tilde{Q}_i^2} - (P_i + \chi R_i)$, so

$$\Delta H_i = H_i^{\text{set}} - H_i^{\text{ap}} = \sqrt{L_i^2 - \tilde{Q}_i^2} - \bar{P}_i + (1 - \chi) R_i. \quad (11)$$

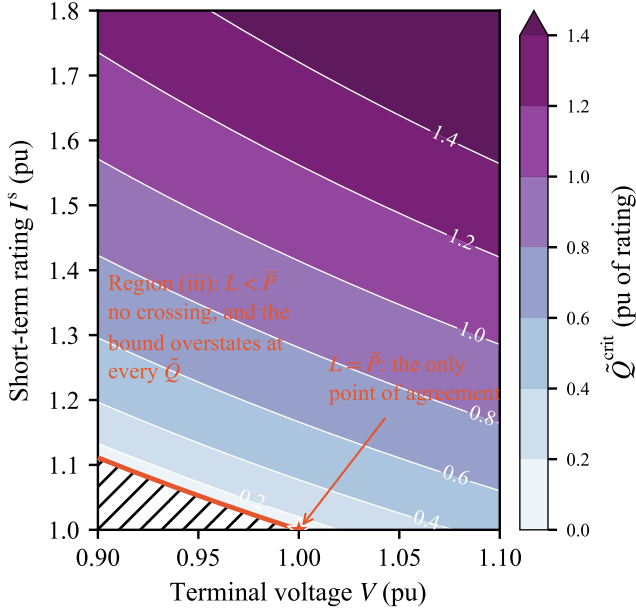


Figure 3: Crossing $\tilde{Q}^{\text{crit}}$ (pu of rating) at $\bar{P} = S$ and full band availability. Below the heavy contour $L < \bar{P}$ and the bound overstates at every reactive dispatch.

At $\chi = 1$ the difference does not depend on the sustained active power.

Proposition. Let $\chi = 1$ and L_i be given by Eq. (10). (i) If $L_i > \bar{P}_i$, define $\tilde{Q}_i^{\text{crit}} = \sqrt{L_i^2 - \bar{P}_i^2}$. The active-power bound understates the inertial headroom for $|\tilde{Q}_i| < \tilde{Q}_i^{\text{crit}}$, is exact at $|\tilde{Q}_i| = \tilde{Q}_i^{\text{crit}}$, and overstates it for $|\tilde{Q}_i| > \tilde{Q}_i^{\text{crit}}$. (ii) If $L_i = \bar{P}_i$, the two agree only at $\tilde{Q}_i = 0$ and the bound overstates elsewhere. (iii) If $L_i < \bar{P}_i$, the bound overstates for every admissible $\tilde{Q}_i$.

The proof follows from Eq. (11). A higher I^s , a higher voltage or a larger band availability increases L_i and moves the bound towards understatement; a larger $|\tilde{Q}_i|$ moves it towards overstatement. The multiplier that would make Eq. (1) exact, $\gamma_i = \sqrt{L_i^2 - \tilde{Q}_i^2} / \bar{P}_i$, varies with voltage, reactive dispatch, short-term rating and band availability, so no constant derating reproduces it. The quantities in the condition are available at the time of clearing: terminal voltage from the state estimator, ratings from equipment data and reactive dispatch from the schedule. Region (iii) applies when $V_i^{\text{sh}} I^s < \bar{P}_i / S_i$ with the band withdrawn, i.e. at low voltage with little overload capability.

Fig. 3 gives the crossing for $\bar{P} = S$ and full band availability. At $I^s = 1.5$ pu the crossing exceeds the rating over the normal voltage band, so the bound is conservative for practical reactive dispatch. At $I^s = 1.1$ pu it lies between 0.30 and 0.58, a range that ordinary reactive duty reaches. The condition was checked against the exact disc at 1,884 points over $V \in [0.90, 1.08]$, $I^s \in [1.0, 1.8]$, three band availabilities and two bounds, and gives the correct sign at all of them.

3. Market clearing formulation

3.1. Objective and requirements

The clearing covers T periods of length Δ with converters $i \in \mathcal{I}$, synchronous units $g \in \mathcal{G}$ and zones $z \in \mathcal{Z}$. The decisions are converter discharge p^d , charge p^c , curtailment c , reactive output Q , reserve R , inertial power H , band availability a and state of charge σ , and synchronous energy p^e and reserve r^e . With e_i^t the available converter injection, the net converter injection is

$$P_i^t = e_i^t + p_i^{d,t} - p_i^{c,t} - c_i^t, \quad (12)$$

and the objective is

$$\min \Delta \sum_t \left[\sum_i (\kappa^e p_i^{d,t} + \kappa^r R_i^t + \kappa^h H_i^t + \kappa^q Q_i^t + \kappa^{\text{cyc}} (p_i^{d,t} + p_i^{c,t})) + \sum_g (\kappa^{\text{ge}} p_g^{e,t} + \kappa^{\text{gr}} r_g^{e,t}) \right], \quad (13)$$

with energy offers in \$/MWh and availability offers in \$/MW/h. The power balance and the reserve, inertia and zonal reactive requirements are

$$\sum_i P_i^t + \sum_g p_g^{e,t} = D^t + \ell^t(\mathbf{x}^t), \quad (14)$$

$$\sum_i R_i^t + \sum_g r_g^{e,t} \geq \rho D^t, \quad (15)$$

$$\sum_i H_i^t \geq P_\omega^{\text{loss}} - \frac{2j_\omega^{\text{max}}}{f_0} \sum_{g \neq \omega} \mathcal{H}_g S_g \quad \forall \omega \in \mathcal{G}, \quad (16)$$

$$\sum_{i \neq \omega} H_i^t \geq P_\omega^{\text{loss}} - \frac{2j_\omega^{\text{max}}}{f_0} \sum_g \mathcal{H}_g S_g \quad \forall \omega \in \mathcal{I}, \quad (17)$$

$$\sum_{i \in \mathcal{I}_z} Q_i^t \geq \varphi Q^{\text{z},t} \quad \forall z \in \mathcal{Z}_I. \quad (18)$$

In Eq. (14), ℓ^t is the linearised loss term of Section 3.3; it is zero in the first (lossless) clearing. The inertia requirement is written once per credible outage ω , and the contribution of the tripped resource is removed from the side on which it appears. The loss P_ω^{loss} is the scheduled active bound of the unit for both fleets, so the clearing cannot reduce its own contingency by de-loading the unit that sets it. The reactive requirement is written for the zones $\mathcal{Z}_I \subseteq \mathcal{Z}$ that contain a converter; the reactive load of the other zones is served by plant outside the clearing. Commitment is fixed.

The store follows

$$\sigma_i^t = \sigma_i^{t-1} - \frac{\Delta p_i^{d,t}}{\eta_d E_i} + \frac{\Delta \eta_c p_i^{c,t}}{E_i}, \quad \sigma_i^T = \sigma_i^0, \quad (19)$$

without a complementarity constraint; simultaneous charge and discharge of a unit does not occur in any reported schedule. Branch flows are limited through the shift-factor matrix Ψ ,

$$|\Psi(\mathbf{P}^t + \mathbf{p}^{g,t} - \mathbf{d}^t)| \leq \mathbf{f}^{\text{max}}, \quad (20)$$

and the net converter injection is bounded by the scheduled active limit, $-P_i^{\text{sch}} \leq P_i^t \leq P_i^{\text{sch}}$.

Table 2 distinguishes the three bounds that appear in the comparison. P_i^{sch} and S_i are common to all models; $\bar{P}_i$ belongs to the comparison model of Eq. (1) only. Results are reported for both $\bar{P}_i = S_i$ and $\bar{P}_i = P_i^{\text{sch}}$.

Table 2: Converter bounds used in the clearing and in the comparison model.

Bound	Quantity limited	Value
P_i^{sch}	net scheduled active injection	rated active power of the converter
S_i	radius of the capability disc	$P_i^{\text{sch}}/(P/S)$, Section 4.1
$\bar{P}_i$	$P_i + R_i + H_i$ in Eq. (1)	S_i or P_i^{sch} (both reported)

3.2. Prices

Prices are the multipliers of the requirements [30]: λ^t for Eq. (14), $\mu^{t,t}$ for Eq. (15) and $\mu^{z,t}$ for Eq. (18). The inertia requirement has one row per outage, and the individual rows are degenerate, so the multiplier of a single row depends on the basis returned by the solver. The inertia price reported here is therefore defined as the marginal cost of raising all inertia rows of a period by one megawatt, i.e. the sum of their multipliers, averaged over the day. This quantity is a directional derivative of the optimal cost. It is verified by re-clearing with perturbations of 1, 0.25 and 0.01 MW, and its left and right limits are reported as a price range. With fixed commitment the clearing is a linear programme, so no pricing rule for integer decisions is needed [31, 32].

3.3. AC-feasible clearing

Equations (20) and (18) do not guarantee that a schedule satisfies the ac network limits. The final schedules are therefore obtained by successive linear programming [33, 34]. Round k consists of four steps.

1. *Clearing.* The linear programme is solved with the rows of step 4 from round $k - 1$. A lexicographic second solve selects, among schedules of equal cost, the one with least storage throughput, which makes the schedule unique; prices are taken from the first solve.
2. *Power flow and voltage control.* An ac power flow is solved for each period with converters as PQ injections. Generator and reference-machine voltage setpoints and transformer taps are adjusted by a least-change search until bus voltages and machine reactive outputs are within limits; reactive limits are not enforced by the solver, so a violation appears as a reactive excess. These controls are not market products and carry no price.
3. *Validation.* The schedule is checked against every bus voltage limit, every machine reactive limit, the loading of every branch at its more heavily loaded end, the difference between the cleared and the actual output of the reference machine together with its active limits and sold reserve, and, for the capability set, each converter current recomputed at the ac terminal voltage.
4. *Linearisation.* Sensitivities of bus voltages, machine reactive outputs, branch flows and losses to the controlled injections are taken about the power-flow solution by central differences. They give (a) the loss term ℓ^t of Eq. (14); (b) branch apparent-power limits as inscribed polygons, tightened by the ratio of measured to linearised loading; (c)

rows that remove small residual voltage and reactive violations or otherwise prevent them from increasing; and (d) a trust region on the active and reactive injection at each controlled bus, which shrinks geometrically from 4 % of peak demand. The converter terminal voltages in Eqs. (8)–(9) are updated.

The iteration stops when the validation of step 3 passes within the tolerances of Table 6. Every subproblem is a linear programme, and the reported prices are the multipliers of the final one, i.e. marginal prices conditional on the final linearisation and control settings.

3.4. Post-contingency deliverability

Equation (20) limits the pre-contingency dispatch only. For a set Ω of secured generator outages, the procured response is also required to reach the network. After the loss of unit ω , its scheduled output $p_\omega^{g,t}$ is replaced by the held response, deployed in proportion $\phi_\omega^t \in [0, 1]$ to the holdings $R_i^t + H_i^t$ and $r_g^{g,t}$ ($g \neq \omega$), and the remainder is supplied by the surviving synchronous machines in proportion to their stored energy $\mathcal{H}_g S_g$. With $\mathbf{s}_\omega$ that sharing vector and $\mathbf{u}_\omega^t$ the vector of deployed holdings,

$$|\Psi(\mathbf{P}^t + \mathbf{p}^{g,t} - \mathbf{d}^t - p_\omega^{g,t} \mathbf{1}_\omega + \phi_\omega^t \mathbf{u}_\omega^t + (p_\omega^{g,t} - \phi_\omega^t \mathbf{1}^\top \mathbf{u}_\omega^t) \mathbf{s}_\omega)| \leq \varepsilon \mathbf{f}^{\max} \quad (21)$$

for all $\omega \in \Omega$ and t , where $\varepsilon = 1.25$ is the short-term emergency rating factor. The fraction $\phi_\omega^t = \min(1, p_\omega^{g,t}/\mathbf{1}^\top \mathbf{u}_\omega^t)$ is a ratio of decisions; it is fixed from the previous solution and iterated with damping until it changes by less than 10^{-3} , so Eq. (21) is linear in each solve.

4. Case study

4.1. Test systems and converter ratings

Two public systems distributed with pandapower [35] are used: the IEEE 39-bus system [36] and the IEEE Reliability Test System (RTS-24) [37]. The first has a few large machines and a network that rarely binds; the second has many small units and a network that binds. All units in the `sgen` and `ext_grid` tables are scheduled, including the reference machine. Both source cases violate their own voltage limits (1.0636 pu against 1.06 pu on IEEE 39-bus; 0.919–1.058 pu against 0.95–1.05 pu on RTS-24). They are corrected before use by moving one generator setpoint on IEEE 39-bus and five setpoints and five transformer taps on RTS-24; active power is not redispatched. After the correction both cases are within their voltage bands, the largest branch loading is 77 % and 90 %, and the change in any reported cost difference is below 0.001 percentage points.

Converters replace the smallest units until a target share of installed capacity is reached (Table 3); Section 5.3 randomises this choice. The rated active power of a converter is that of the unit it replaces, $P_i^{\text{sch}} = P_{\max,i}$. Its apparent-power rating is defined by the ratio P/S . Three ratios with a direct interpretation are used: $P/S = 1.00$, an inverter sized to its active power with no reactive capability at full output; $P/S = 0.95$, the power-factor capability commonly required of inverter-based plant at

Table 3: Test systems. Converter share of installed capacity is given as target / achieved.

System	Share	Converters		Synchronous	
		no.	MVA	no.	MVA
IEEE 39-bus	0.2 / 0.225	3	1779	7	6115
peak 5,894 MW	0.4 / 0.404	5	3190	5	4704
RTS-24	0.2 / 0.212	19	798	13	2961
peak 2,724 MW	0.4 / 0.445	25	1671	7	2088

rated active power; and $P/S = 0.90$, the stricter requirement, close to a storage inverter sized for four-quadrant operation.

The reference convention takes $S_i = \sqrt{P_{\max,i}^2 + Q_{\max,i}^2}$ from the reactive limits of the replaced unit, which gives P/S between 0.857 and 0.952 (0.909–0.929 on average by configuration), inside the range of the other three. Cost and price results are reported for all four conventions and for both comparison bounds (Section 5.2). DC-side limits of the converter and the store are not modelled; the terminal capability represents deliverability when the dc source can supply $P_i + R_i$ for the containment period and $P_i + R_i + H_i$ for the short-term window, which Eq. (7) enforces in energy but not in power.

4.2. Parameters and comparison models

The day consists of eight three-hour periods with load factors 0.72, 0.68, 0.80, 0.95, 1.00, 0.97, 1.05 and 0.85; the peak is 80 % of the installed active capacity and is the same at both converter shares. Offers are illustrative. A factor common to all offers does not affect the comparisons, but the ratio of availability offers to energy offers does, and it is varied in Section 5.3. Problems are solved with the HiGHS dual simplex through SciPy [38, 39].

The capability set is compared with three models that share all other constraints: (i) the active-power bound of Eq. (1); (ii) an apparent-power model, $\|(P_i + R_i + H_i, Q_i)\|_2 \leq S_i$, represented by the same inscribed m -gon, without terminal voltage or short-term rating; and (iii) a uniform derating of the active-power bound, $|P_i + R_i + H_i| \leq \gamma^{\text{UD}} S_i$ with $|Q_i| \leq \kappa^{\text{Q}} S_i$, where κ^{Q} is the smallest per-unit reactive range that allows every zonal requirement to be met and $\gamma^{\text{UD}} = [(V^{\min} I^c)^2 - (\kappa^{\text{Q}})^2]^{1/2}$, so that the continuous current limit holds at every voltage of the day. Differences are reported as

$$\delta = 100(C^{\text{cmp}} - C^{\text{set}})/C^{\text{cmp}} \%, \quad (22)$$

with C^{cmp} the cost under the comparison model and C^{set} that under the capability set; $\delta > 0$ means the comparison model is the more expensive. With all three omissions removed ($I^s = I^c$, no reactive requirement, $V = 1.0$ pu) the capability set and Eq. (1) return the same prices and costs within 0.0023 %. The Supplementary Material gives the variable domains, the state-of-charge gate and polygon-accuracy checks, the price-range computation, the reduction test, the full parameter sweep, the IEEE 118-bus check, the post-contingency screen, the RMS converter model, the crossing values of the Proposition, the source-case corrections and the parameters (Sections S1–S10).

Table 4: δ (%) by converter rating convention and comparison bound ($e = 0.8$ pu, $I^s = 1.5$ pu).

P/S	$\bar{P}$	IEEE 39		RTS-24	
		0.2	0.4	0.2	0.4
1.00	$S = P^{\text{sch}}$	1.02	0.53	0.62	5.44
0.95	S	0.98	0.41	0.67	2.67
0.95	P^{sch}	2.83	1.14	1.02	6.89
0.90	S	0.93	0.26	0.56	1.07
0.90	P^{sch}	4.85	4.49	1.67	8.25
reference	S	0.96	0.34	0.68	1.08
reference	P^{sch}	3.67	1.90	1.40	7.94

Table 5: δ (%) against the active-power bound (AP), apparent-power model (S) and uniform derating (UD) at two dispatch levels.

System, share	$e = 0.8$ pu			$e = 1.0$ pu		
	AP	S	UD	AP	S	UD
IEEE 39, 0.2	0.96	1.01	0.93	9.00	9.38	8.82
IEEE 39, 0.4	0.34	0.61	2.04	14.87	16.74	22.49
RTS-24, 0.2	0.68	0.73	0.58	1.17	1.60	0.65
RTS-24, 0.4	1.08	1.30	1.69	1.92	3.29	5.32

5. Results

5.1. Sign condition in the clearing

Fig. 4 varies each omitted quantity with the other two removed. The short-term rating gives $\delta \geq 0$, up to 1.1 %: the bound cannot use the overload band. Reactive duty gives $\delta \leq 0$, down to -2.6 %: the bound does not account for reactive current; at a requirement of 120 % of zonal reactive load neither representation clears at 40 % converter share, because the requirement exceeds the reactive rating of the converters in the binding zone, so that point is absent from Fig. 4c. Terminal voltage gives a δ that changes sign at 1.0 pu, from -2.7 % at 0.95 pu to 0.9 % at 1.05 pu. Each response follows the Proposition: a larger L_i raises $\bar{Q}_i^{\text{crit}}$ and moves δ upwards, and a larger $\bar{Q}_i$ moves it downwards. The signs are the same in all four configurations; the magnitudes depend on the system and on how heavily the converters are loaded, and do not increase monotonically with the converter share. Because the three responses differ in sign, and the voltage response changes sign inside the normal operating band, a constant derating or uplift of Eq. (1) cannot reproduce the capability set.

5.2. Cost and inertia price

Table 4 gives δ for the four rating conventions and the two comparison bounds. The sign is positive in all 28 cells (the two bounds coincide at $P/S = 1.00$). The magnitude depends on the convention: from 0.26–5.44 % with $\bar{P} = S$ and from 0.53–8.25 % with $\bar{P} = P^{\text{sch}}$, the $P/S = 1.00$ row counting for both. The sign condition and the direction of the capability mismatch are therefore properties of the two feasible sets, whereas the size of the cost difference is conditional on the rating assumption and should be read with it.

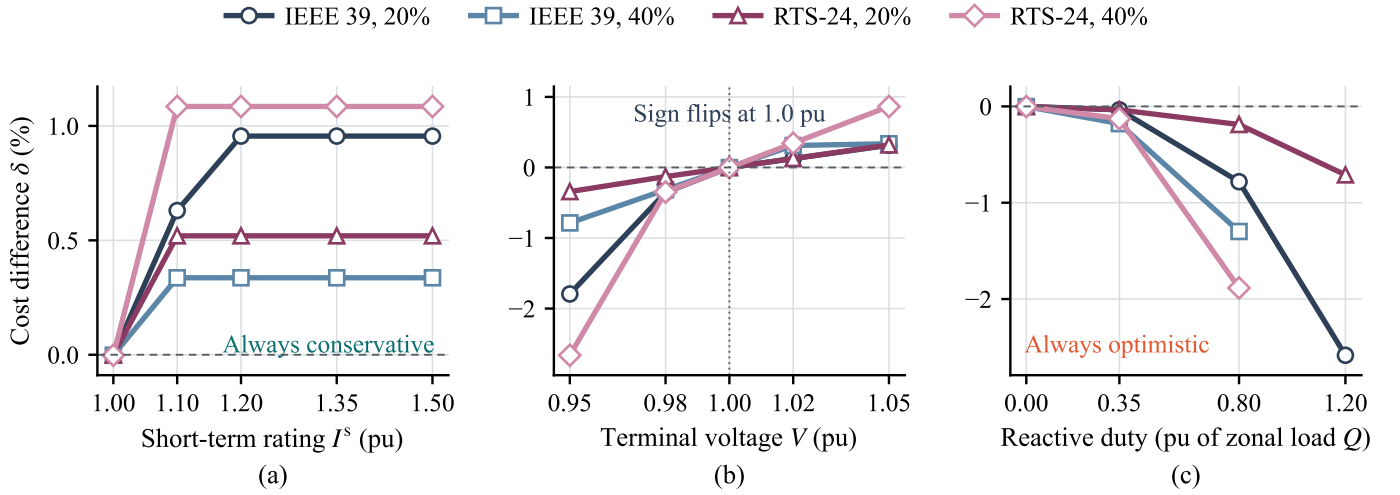


Figure 4: δ for each omitted quantity with the other two removed ($e = 0.8$ pu, $m = 192$): (a) short-term rating, (b) terminal voltage, (c) reactive requirement as a fraction of zonal reactive load.

The inertia price under the capability set stays within 0.1 % of the converter offer, 3.0 \$/MW/h, in every configuration, rating convention and dispatch level: with the short-term band available, inertial power is not scarce. Under Eq. (1) inertial power competes with energy for the same megawatts, and the price rises with converter loading (Fig. 5a). At $e = 0.8$ pu it is 6.75–8.0 with $\bar{P} = S$, a ratio of 2.3–2.7, and 25.6–41.0 with $\bar{P} = P^{\text{sch}}$ (reference rating); at $e = 1.0$ pu with $\bar{P} = S$ the ratio is 7.8–13.7. For both models the left and right limits of the price coincide at perturbations of 1, 0.25 and 0.01 MW, so the price ranges do not overlap and the ratio is determined; it remains conditional on the illustrative offers, and only its direction is general. The cost difference also grows with loading: it is zero at $e = 0.4$ pu and, at $e = 1.0$ pu, the bound clears at 1.175 times the capability-set cost on IEEE 39-bus (40 % share) and 1.020 times on RTS-24, mainly through replacement energy for curtailed converter output (Fig. 5b). Curtailment is a primal quantity on a wide optimal face, so it is reported as a range over that face: on IEEE 39-bus the bound curtails at least 7,243 MWh more than the capability set at every optimum, while on RTS-24 the two ranges overlap and no difference is claimed.

Fig. 6 follows RTS-24 at 40 % share through the day. From the third period the capability set holds at least 43 MW more converter reserve than the bound, and from the fourth period between 56.5 MW and 110.4 MW more; these are differences between the ranges over the two optimal faces, so they hold at every optimum. Under the capability set the converter reserve is unique in every period, whereas under the bound its range is 53–60 MW wide from the third period on. The state-of-charge trajectories coincide, so the difference is not caused by storage use.

5.3. Comparison models, approximation error and generality

Table 5 gives δ against the three comparison models. No comparison model is cheaper than the capability set in any

of the sixteen combinations of configuration and dispatch; the difference is zero at light loading and positive otherwise. At full dispatch it is 1.2–14.9 % against the active-power bound, 1.6–16.7 % against the apparent-power model and 0.6–22.5 % against the uniform derating. The uniform derating keeps every converter within its continuous current, as the capability set does; the active-power bound does not, and its lossless schedules reach 1.38 pu of continuous current on RTS-24, and 1.34–1.40 pu after the ac loop (Table 6, last column). The exact second-order cone formulation gives δ within 0.056 percentage points of the polygon value, and the polygon value is the smaller in every case, so the reported differences are conservative.

Fig. 7 summarises the sensitivity of δ to the parameters that a decarbonising system changes. A tighter RoCoF limit, a lower synchronous inertia constant and a shorter storage duration all increase δ ; at a RoCoF limit of 2 Hz s^{-1} the inertia requirement no longer binds on IEEE 39-bus and the two representations coincide. The offer ratio scales δ almost proportionally without changing its sign. The remaining parameters (polygon faces, cycling cost, reserve overlap, state-of-charge thresholds, terminal-voltage offset) change δ by at most 0.4 percentage points (0.41 at $m = 8$ on RTS-24, 20 % share, where the octagon withholds 10.8 % of the active coordinate).

To test whether the result depends on which units are converters, the selection was randomised: units are drawn in random order until the capacity share of the reference placement is reached within 2 %. Of 666 accepted draws, 605 have a feasible clearing, and $\delta > 0$ in all of them (Fig. 8). Within a configuration the magnitude varies by a factor of 2.6–34.5, and the reference placement lies in the lower part of the distribution in three of the four configurations. The 61 infeasible draws are all on RTS-24 and are infeasible under both representations: the zonal reactive requirement cannot be met with those converter locations. On the IEEE 118-bus system, under the dc approxima-

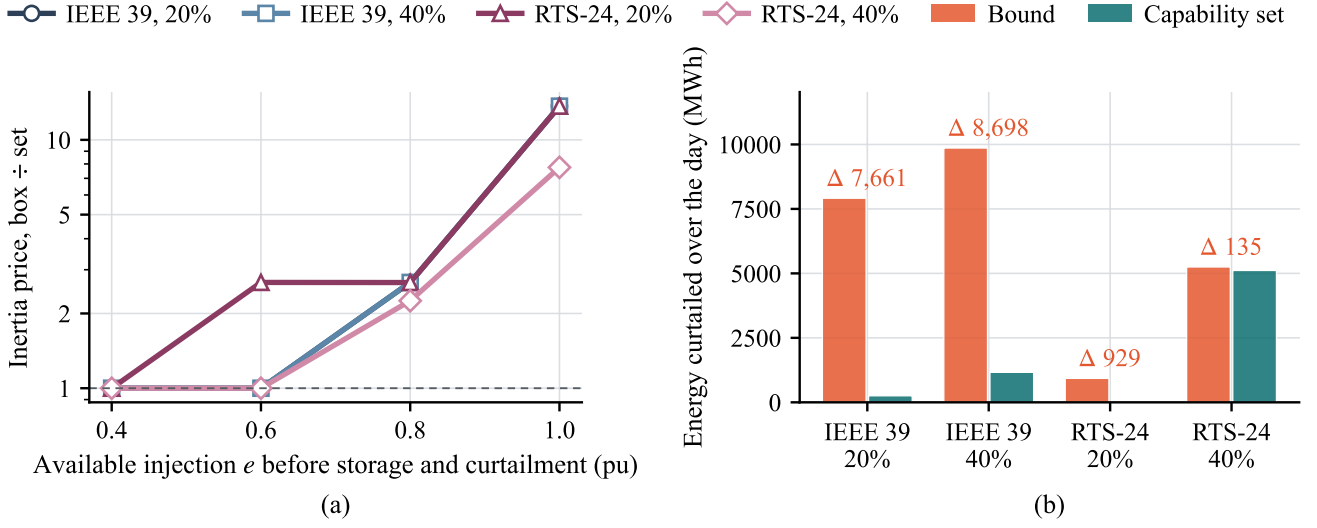


Figure 5: (a) Ratio of the inertia price under the active-power bound ($\bar{P} = S$) to that under the capability set against available converter injection. (b) Curtailed converter output at $e = 1.0$ pu at the selected optimum.

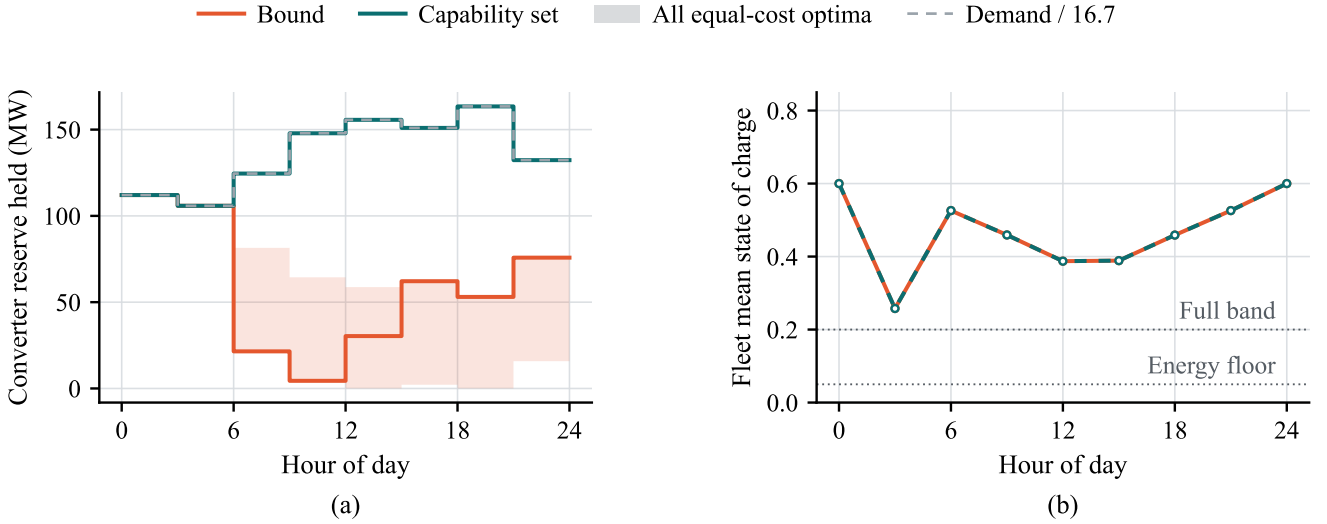


Figure 6: RTS-24, 40 % converter share: (a) converter reserve, selected optimum (line) and range over the optimal face (band); (b) fleet-mean state of charge.

tion and with $P/S = 0.95$, the inertia requirement binds only at 40 % share with a RoCoF limit of 0.5 Hz s^{-1} ; there $\delta = 0.12 \%$ at full dispatch (zero at $e = 0.8$ pu) and the price ratio is 2.7. Where it does not bind, only the reactive-current omission acts, and δ is zero at $e = 0.8$ pu and -0.68% at full dispatch, the sign of Fig. 4c. The direction of the error on a given system thus follows from which constraint binds, as the sign condition states.

5.4. AC validation of the final schedules

The procedure of Section 3.3 was applied to the capability set and to the active-power bound in all four configurations. All

eight final schedules satisfy the bus voltage, machine reactive, branch loading and balance limits in an ac power flow (Table 6), in 5–13 rounds. The first, lossless clearing leaves 33–76 MW for the reference machine to supply and loads a branch to at most 103.2 %; in the final schedules the reference machine deviates from its cleared output by less than 1 MW and the largest branch loading is 99.9 %. Charging the schedule for its losses raises the cost by 1.0–2.4 %. The comparison is unchanged in sign: δ is 0.85, 0.07, 0.59 and 1.00 % for the ac-feasible schedules against 0.96, 0.34, 0.68 and 1.08 % for the lossless ones, and the inertia price is 3.0–3.03 under the capability set and 6.8–8.5 under the bound. With the converter currents recom-

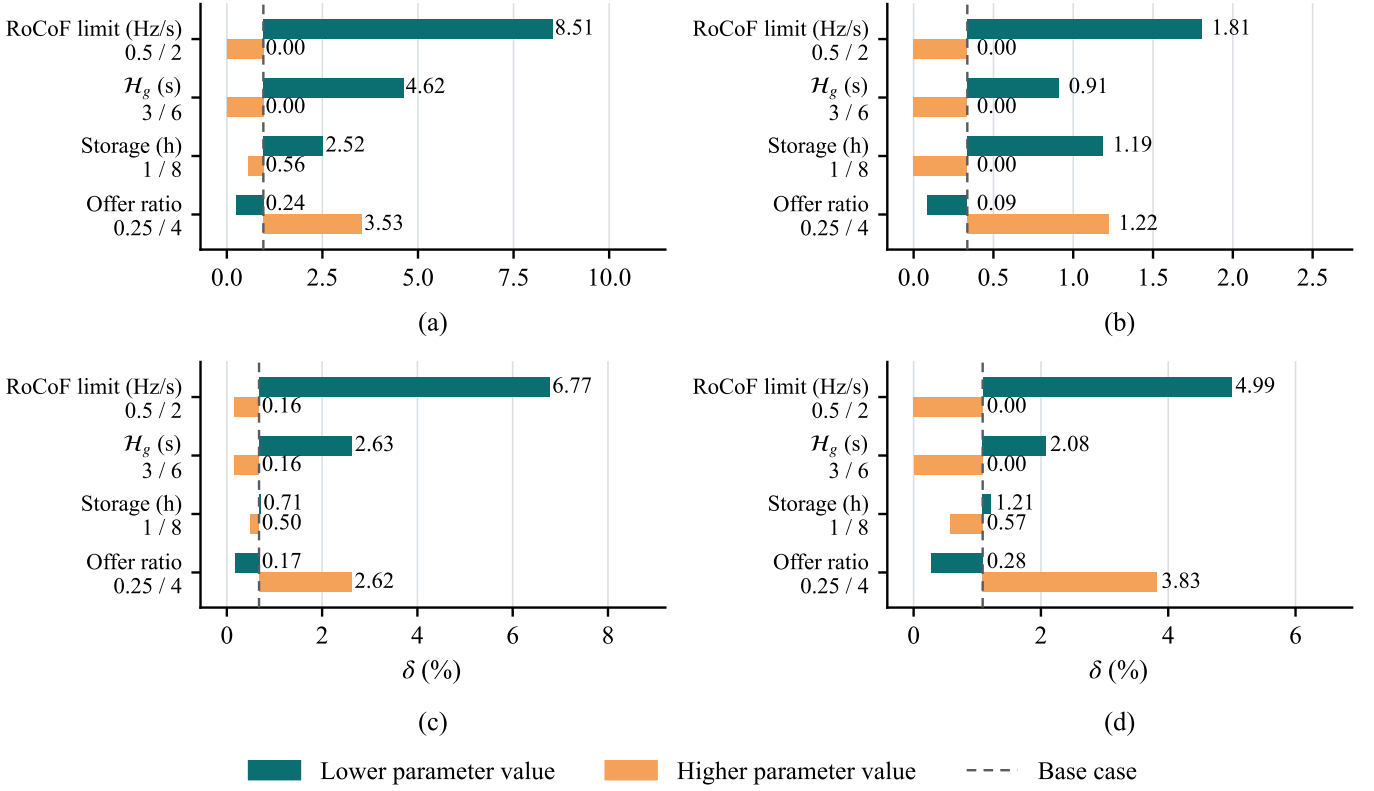


Figure 7: Sensitivity of δ to the RoCoF limit, synchronous inertia constant, storage duration and offer ratio ($\bar{P} = S$); bars run from the base case (dashed) to δ at the lower and higher parameter value given under each label. (a) IEEE 39-bus, 20 % share, base $\delta = 0.96$ %; (b) IEEE 39-bus, 40 %, 0.34 %; (c) RTS-24, 20 %, 0.68 %; (d) RTS-24, 40 %, 1.08 %.

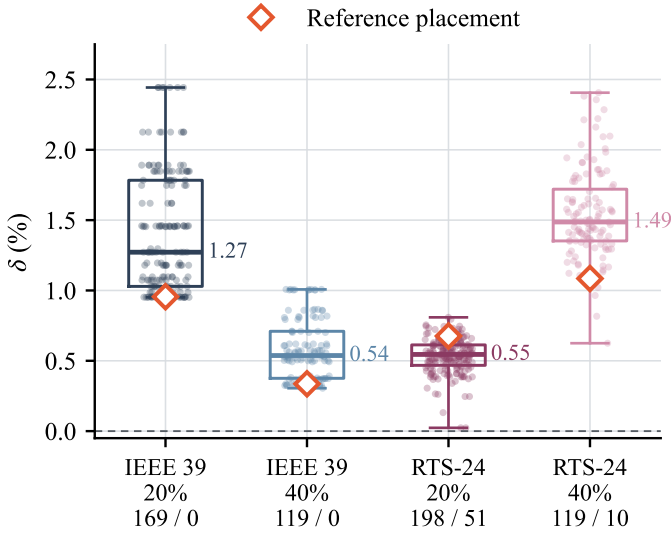


Figure 8: δ over the randomised converter placements: box, quartiles; whiskers, range; printed value, median; numbers under each configuration, feasible / infeasible draws.

puted at the ac terminal voltages, the capability-set schedules stay at or below the continuous rating, whereas the schedules of the bound reach 1.34–1.40 pu on RTS-24.

5.5. Post-contingency deliverability

Equation (21) was imposed for the eight synchronous units with the largest active limits, or for all of them where fewer are committed (five to eight per configuration). Without it, the response model of Section 3.4 overloads a branch beyond its emergency rating in up to 29 of 47 secured outage-period cases, with a worst loading of 129 %. With it, the worst loading is within 0.03 % of the emergency rating in the six clearings whose screen had exceeded it, and unchanged at 97.0 and 90.1 % on RTS-24 at 20 % share, where it had not. The cleared cost rises by at most 0.06 %, the inertia and reserve prices are unchanged, and δ is unchanged except on RTS-24 at 40 % share, where it moves from 1.08 % to 1.14 % (Table 7). The results of Sections 5.2 and 5.3 therefore hold when the procured response is required to be deliverable after the critical outages. Outages outside the secured set are screened but not constrained.

5.6. Dynamic verification of the short-term constraint

Whether a converter delivers the scheduled inertial power within its current limit was tested with an RMS model of one GFM converter: a virtual synchronous machine behind a filter reactance, connected through a reactance x^g to a Thévenin source, with a current limiter that applies the two ratings and the priority rule of Eq. (4) and an anti-windup clamp (Supplementary Section S8). The virtual inertia is set so that the scheduled H_i is requested at the design RoCoF of 1 Hz s^{-1} . The 360

Table 6: AC power-flow validation of the final schedules. Tolerances: voltage 10^{-4} pu, loading 0.05 %, reactive excess 0.5 MVar, balance 1 MW, converter current 10^{-3} pu.

System, share	Model	Rounds	$V^{\min}$ (pu)	$V^{\max}$ (pu)	Loading (%)	Q exc. (MVar)	Balance (MW)	Δ cost (%)	I^{conv} (pu)
IEEE 39, 0.2	bound	5	0.9432	1.0600	98.6	0.0	0.96	+1.05	0.83
IEEE 39, 0.2	set	10	0.9400	1.0600	99.5	0.0	0.03	+1.16	1.00
IEEE 39, 0.4	bound	6	0.9401	1.0600	94.8	0.0	0.37	+1.30	1.06
IEEE 39, 0.4	set	6	0.9400	1.0600	95.2	0.0	0.55	+1.57	1.00
RTS-24, 0.2	bound	6	0.9650	1.0500	86.7	0.0	0.64	+1.38	1.34
RTS-24, 0.2	set	9	0.9588	1.0500	97.3	0.0	0.05	+1.47	1.00
RTS-24, 0.4	bound	5	0.9556	1.0500	99.7	0.0	0.20	+2.31	1.40
RTS-24, 0.4	set	13	0.9558	1.0481	99.9	0.0	0.00	+2.40	1.00

Table 7: Post-contingency deliverability for the secured generator outages: worst branch loading relative to the emergency rating without and with Eq. (21), and cost increase.

System, share	Model	$ \Omega $	Worst loading (%)		Cost (%)
			without	with	
IEEE 39, 0.2	bound	7	113.9	100.0	0.00
IEEE 39, 0.2	set	7	129.2	100.0	0.00
IEEE 39, 0.4	bound	5	109.2	100.0	0.00
IEEE 39, 0.4	set	5	122.9	100.0	0.00
RTS-24, 0.2	bound	8	97.0	97.0	0.00
RTS-24, 0.2	set	8	90.1	90.1	0.00
RTS-24, 0.4	bound	7	105.6	100.0	0.06
RTS-24, 0.4	set	7	107.6	100.0	0.00

runs combine $V \in \{0.95, 1.00, 1.05\}$ pu, low and high reactive schedules and the crossing $\tilde{Q}^{\text{crit}}$ (conservative, crossing and optimistic regions of the Proposition), $I^s \in \{1.1, 1.5\}$ pu, reactive and active priority, $x^g \in \{0.05, 0.15, 0.30\}$ pu, and a grid voltage that is either constant or falls by 5 % at the event. At $I^s = 1.5$ pu the crossing exceeds the rating, so only the conservative region occurs there; each reactive mode and grid-voltage case therefore comprises 90 runs.

The limiter becomes active in all runs, 0.07–0.34 s after the event, and the converter current does not exceed the limit in force (Fig. 9, Table 8). In every run the settled active power equals the boundary of Eq. (4) evaluated at the terminal voltage reached during the event. The shortfall against the schedule is therefore a voltage effect. With the grid voltage constant, the terminal voltage falls by 0.06–0.25 % for $x^g = 0.05$ pu and the converter delivers 99.5–99.7 % of the scheduled inertial power; for $x^g = 0.15$ pu the figures are 0.5–2.3 % and 95.1–97.7 %; for $x^g = 0.30$ pu the voltage falls by up to 9.5 % and delivery is 79.9 % in the worst case. With the grid voltage falling by 5 % the worst case is 54.0 %. Delivery correlates with x^g (−0.93) and not with the region of the Proposition. Under voltage-droop reactive control the converter holds its terminal voltage at the expense of active current, and in 16 of 90 droop runs with falling grid voltage the active response is lost; the constraint assumes that the reactive schedule is held, as Eq. (5) states.

The pre-contingency voltage is thus not a safe value for the short-term constraint at weak buses, and the constraint is corrected rather than its coefficients. With the reactive schedule

Table 8: RMS verification with the reactive schedule held and constant grid voltage: delivered inertial power (% of schedule), terminal-voltage drop and time of limiter activation.

Region (I^s)	x^g (pu)	Delivered (%)	ΔV (%)	t^{lim} (s)
Conservative (1.1, 1.5 pu)	0.05	99.5–99.7	0.1–0.3	0.12–0.17
	0.15	95.1–97.7	0.9–2.3	0.17–0.25
	0.30	79.9–90.6	3.5–9.5	0.23–0.34
Crossing (1.1 pu)	0.05	99.6–99.7	0.1	0.10–0.11
	0.15	96.8–97.4	0.6–0.9	0.15–0.17
	0.30	86.9–89.3	2.4–3.6	0.21–0.23
Optimistic (1.1 pu)	0.05	99.5–99.7	0.1	0.08–0.10
	0.15	95.8–97.3	0.5–0.6	0.12–0.15
	0.30	82.9–89.1	2.1–2.3	0.17–0.21

held, the terminal voltage during the event follows from the reactance behind the terminal,

$$(V_i^{\text{sh}})^2 = V_i^2 - (x_i^g)^2 [(I^s)^2 - I_i^2], \quad (23)$$

with I_i the pre-event current; it reproduces the simulated voltages to numerical precision. The reactance x_i^g is obtained from the voltage sensitivities of Section 3.3. Using V_i^{sh} in Eq. (4) lowers the short-term face by up to 13.0 %, a terminal voltage of 0.88 pu against a pre-event 1.01 pu (IEEE 39-bus, 40 % share). Re-clearing with the corrected face leaves the cost and the total scheduled inertial power unchanged and reallocates the inertial obligation between converters. When the corrected schedules are passed through the converter model, the energy delivered in the containment window, weighted by scheduled inertial power, changes from 86.3 to 89.9 % and from 80.7 to 83.4 % on IEEE 39-bus, and from 92.8 to 93.3 % and from 97.7 to 97.0 % on RTS-24; the tenth percentile rises from 54.0 to 75.9 % where it was lowest. The remaining shortfall is the rise time of the response, which depends on the virtual inertia and damping and not on the capability set.

6. Conclusion

The active-power bound used in shared-capability scheduling is the projection of the converter current limit onto zero reactive power at unit voltage without a short-term rating. This paper replaced it by a polyhedral capability set with continuous and short-term current ratings, reactive current and terminal voltage, in a multi-period linear clearing of energy, reserve

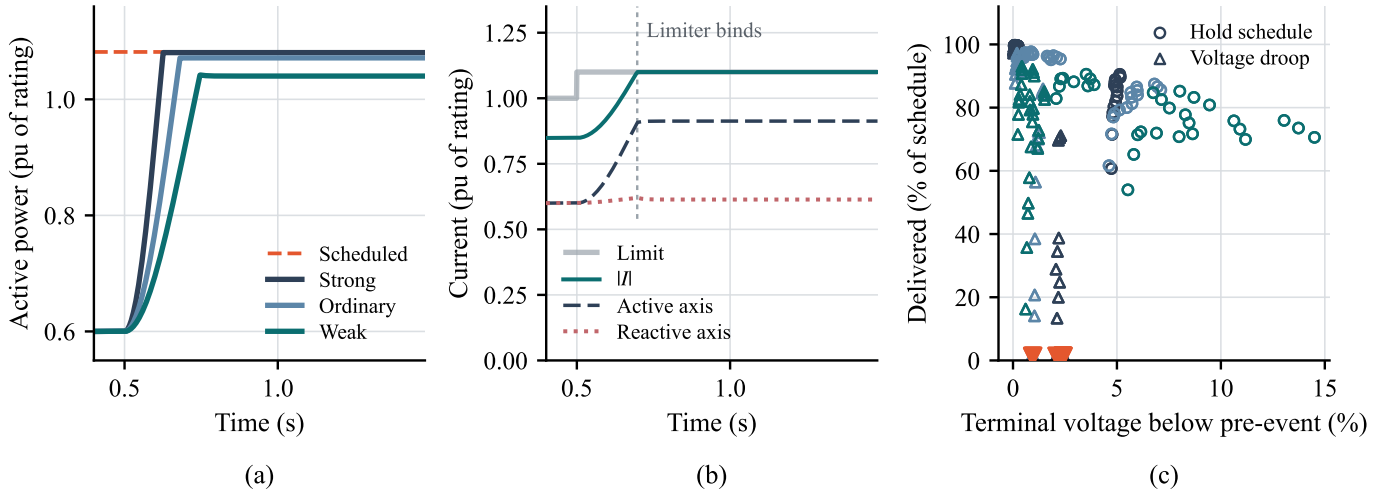


Figure 9: RMS verification: (a) active power for three grid reactances against the schedule; (b) current magnitude, limit and axis currents for $x^g = 0.30$ pu; (c) delivered against scheduled inertial power versus terminal-voltage drop, all 360 runs.

and inertia, and derived a piecewise condition for the sign of the error of the bound. The condition depends on the short-term radius, the bound and the reactive current held during the response. In the clearing, the three omissions produce errors of different sign (0–1.1 %, –2.7–0.9 % and –2.6–0 % of cost), so the bound cannot be corrected by a constant factor.

On the IEEE 39-bus and RTS-24 systems the capability set was never more expensive than the active-power bound, the apparent-power model or a safe uniform derating, in 28 rating and bound combinations and 605 randomised placements. The cost difference is 0.3–1.1 % at the base case, up to 8.5 % with a tighter RoCoF limit and 0.6–22.5 % at full converter dispatch; its size depends on the rating convention and on the offers. The capability set prices inertial power at the converter offer, while the bound prices it 2.3–13.7 times higher depending on loading and bound. All final schedules satisfy voltage, machine reactive, branch and balance limits in an ac power flow, the results hold when the response is constrained to be deliverable after the largest generator outages, and an RMS converter model delivers the scheduled inertial power within the current limit at the terminal voltage reached during the event, for which a closed-form correction of the short-term constraint was given.

The limitations fall into four groups. *Network and ac feasibility*: the final schedules are ac-feasible for the pre-contingency state, and prices are those of the final linearised subproblem; post-contingency security covers generator outages with a shift-factor model and an emergency rating, not branch outages or post-contingency voltages. *Converter dynamics*: the verification uses one converter in RMS without inner loops, dc-link dynamics, fault current or ride-through logic, and corrected short-term voltages as low as 0.88 pu would engage such functions; an electromagnetic-transient study on a network is needed. *Market and pricing assumptions*: commitment is fixed, offers are illustrative and identical within a technology, and the contingency is taken at the scheduled bound of each unit; no absolute price should be read from the results.

Scale and data: the study uses two benchmark systems and an additional 118-bus check; behaviour on an operator model, optimised siting and dc-side power limits remain for further work.

CRedit authorship contribution statement

Junseok Lim: Conceptualization, Methodology, Software, Validation, Formal analysis, Writing – original draft. **Sanghyuk Ji**: Validation, Investigation, Writing – review & editing. **Byungjun Lee**: Validation, Visualization, Writing – review & editing. **Sungwoo Bae**: Supervision, Funding acquisition, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The code, the modified test cases and converter mapping, the parameters, the result files behind every table and figure, and the scripts that regenerate them are available at <https://github.com/Junseok-Lim-HY/gfm-capability-set-clearing>.

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